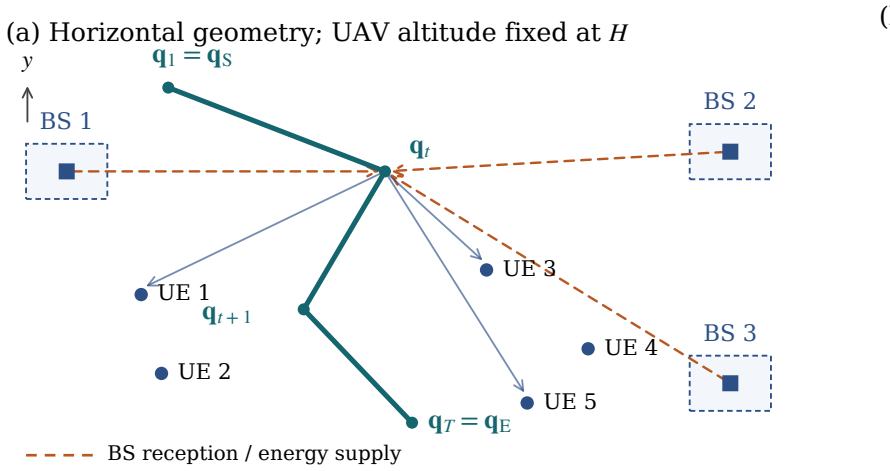
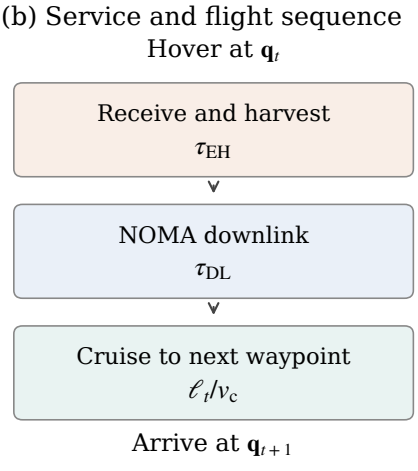

(a) Horizontal geometry; UAV altitude fixed at H
y
$\mathbf{q}_1 = \mathbf{q}_S$
BS 1
BS 2
BS 3
$\mathbf{q}_t$
UE 1
$\mathbf{q}_{t+1}$
UE 2
UE 3
UE 4
UE 5
$\mathbf{q}_T = \mathbf{q}_E$
BS reception / energy supply


(b) Service and flight sequence
Hover at $\mathbf{q}_t$
Receive and harvest
τ_{EH}
NOMA downlink
τ_{DL}
Cruise to next waypoint
ℓ_t / v_c
Arrive at $\mathbf{q}_{t+1}$